



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For



Security Operations for Autonomous AI Agents

TGS Ref No: TGS-2024042604

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 September 2026	Initial 4-day lesson plan aligned to the 454-slide technical deck and 10 activities.	Dr Alfred Ang

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Course Information

Course title	Security Operations for Autonomous AI Agents
Course code	TGS-2024042604
TSC	Security Strategy · ICT-SNA-5021-1.1
Duration	4 days · 32 scheduled hours, excluding lunch
Mode	Instructor-led classroom or synchronous delivery with offline activities
Assessment	WA 1 hour + PP 3 hours, matching the verified legacy papers

Learning Outcomes

- LO1: Formulate comprehensive security goals and establish business standards with an overarching security vision for autonomous AI agents.
- LO2: Communicate effective agent-security policies and practices, and manage compliance with best practices and technological advancements.
- LO3: Evaluate existing agent-security controls against business risks and costs, and develop strategies to resolve identified security gaps.
- LO4: Implement organisation-wide agent-security initiatives, assessing and addressing the impact of security gaps.

Four-Day Schedule

Day 1 — Security Strategy and Standards for Autonomous AI Agents

Time	Type	Learning activity	Evidence / slide reference	Duration
09:30–10:00	Admin	Welcome, attendance, outcomes and assessment orientation	Slides 1–18	30 min
10:00–11:00	Topic	Strategy operating model and criteria	Slides 19–127	60 min
11:00–11:15	Break	Tea break	—	15 min
11:15–13:30	Technical	Agent inventory, goals, identity, tools and data controls	Topic 1 slides	135 min
13:30–14:30	Lunch	Lunch break	—	60 min
14:30–16:00	Activity	Activities 01–02: inventory and trust-boundary threat model	Activity folders 01–02	90 min
16:00–16:15	Break	Tea break	—	15 min
16:15–18:15	Activity	Activity 03: enforce tool and data guardrails	Activity folder 03	120 min
18:15–18:30	Recap	Evidence review and PM attendance	Topic 1 recap	15 min

Scheduled time excluding lunch: 480 minutes (8.0 hours).

Day 2 — Agent Security Policies, Governance and Compliance

Time	Type	Learning activity	Evidence / slide reference	Duration
09:30–09:45	Recap	Day 1 recap and AM attendance	Topic 1 recap	15 min
09:45–11:00	Topic	Policy, governance and compliance operating model	Slides 128–233	75 min
11:00–11:15	Break	Tea break	—	15 min
11:15–13:30	Technical	Policy hierarchy, policy-as-code, permissions and evidence	Topic 2 slides	135 min
13:30–14:30	Lunch	Lunch break	—	60 min
14:30–16:00	Activity	Activity 04: compliance evidence pack	Activity folder 04	90 min
16:00–16:15	Break	Tea break	—	15 min
16:15–18:15	Activity	Activity 05: security posture communication	Activity folder 05	120 min
18:15–18:30	Recap	Governance decisions and PM attendance	Topic 2 recap	15 min

Scheduled time excluding lunch: 480 minutes (8.0 hours).

Day 3 — AI Agent Risk Assessment and Security Control Evaluation

Time	Type	Learning activity	Evidence / slide reference	Duration
09:30–09:45	Recap	Day 2 recap and AM attendance	Topic 2 recap	15 min
09:45–11:00	Topic	Agent threat and risk evaluation operating model	Slides 234–339	75 min
11:00–11:15	Break	Tea break	—	15 min
11:15–13:30	Technical	Injection, poisoning, excessive agency, exfiltration and supply chain	Topic 3 slides	135 min
13:30–14:30	Lunch	Lunch break	—	60 min
14:30–16:00	Activity	Activity 06: quantify risks and control value	Activity folder 06	90 min
16:00–16:15	Break	Tea break	—	15 min
16:15–18:15	Activity	Activity 07: prioritise remediation roadmap	Activity folder 07	120 min
18:15–18:30	Recap	Risk decisions and PM attendance	Topic 3 recap	15 min

Scheduled time excluding lunch: 480 minutes (8.0 hours).

Day 4 — Implementing and Monitoring Organisation-Wide Agent Security Operations

Time	Type	Learning activity	Evidence / slide reference	Duration
09:30–09:45	Recap	Day 3 recap and AM attendance	Topic 3 recap	15 min
09:45–10:45	Topic	Monitoring, detection and	Slides 340–448	60 min

		response operating model		
10:45–11:00	Break	Tea break	—	15 min
11:00–12:00	Activity	Activity 08: suspicious behaviour detection	Activity folder 08	60 min
12:00–12:45	Activity	Activity 09: prompt-injection containment	Activity folder 09	45 min
12:45–13:30	Activity	Activity 10: recovery and post-incident review	Activity folder 10	45 min
13:30–14:30	Lunch	Lunch break	—	60 min
14:30–15:30	Assessment	Written Assessment (WA), open book	7 questions · K1–K7 · 1 hour	60 min
15:30–18:30	Assessment	Practical Performance (PP)	6 tasks · A1–A7 · 3 hours	180 min

Scheduled time excluding lunch: 480 minutes (8.0 hours).

Deck and Activity Alignment

Topic	Criteria	Deck slides	Activities
01 — Security Strategy and Standards for Autonomous AI Agents	K1, K2, A1, A2	19–127	Activity 01, Activity 02, Activity 03
02 — Agent Security Policies, Governance and Compliance	K3, K4, A3, A4	128–233	Activity 04, Activity 05
03 — AI Agent Risk Assessment and Security Control Evaluation	K5, K6, A5, A6	234–339	Activity 06, Activity 07
04 — Implementing and Monitoring Organisation-Wide Agent Security Operations	K7, A7	340–448	Activity 08, Activity 09, Activity 10